AI-Powered Zomato Restaurant Analytics & Rating Prediction System

1. Problem Statement

The restaurant industry generates large amounts of customer and operational data. However, restaurant owners often struggle to identify the factors that influence customer satisfaction and ratings. Understanding customer preferences, delivery services, pricing strategies, and restaurant features is essential for improving business performance. This project aims to analyse the Zomato restaurant dataset to discover patterns affecting restaurant ratings and customer engagement. Additionally, a machine learning model is developed to predict restaurant ratings based on key business features.

2. Objectives

The main objectives of this project are:

1. To clean and preprocess the Zomato dataset.
2. To perform Exploratory Data Analysis (EDA) and identify hidden patterns.
3. To visualize restaurant trends using professional charts and graphs.
4. To analyse the impact of online delivery, table booking, pricing, and customer votes on restaurant ratings.
5. To build a machine learning model capable of predicting restaurant ratings.
6. To provide business recommendations based on data-driven insights.

3. Project Requirements

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or higher
- Storage: 1 GB free space
- Internet Connection

Software Requirements

- Python 3.x
- Jupyter Notebook / Google Colab
- GitHub Account

Libraries Used

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

4. Dataset Description

The dataset was obtained from Zomato and contains restaurant-related information.

Dataset Attributes

- Restaurant ID
- Restaurant Name
- Country Code
- City
- Address
- Locality
- Cuisines
- Average Cost for Two
- Currency
- Has Table Booking
- Has Online Delivery
- Price Range
- Aggregate Rating
- Rating Text
- Votes

The dataset contains information about restaurant services, customer engagement, pricing, and ratings.

5. Methodology

The project was completed in the following stages:

Step 1: Data Collection

The Zomato dataset was imported into Python using Pandas.

Step 2: Data Cleaning

- Removed duplicate records
- Handled missing values
- Checked data types
- Prepared data for analysis

Step 3: Exploratory Data Analysis

The following questions were analysed:

1. Does online delivery affect restaurant ratings?
2. Does table booking influence customer satisfaction?
3. Which cities have the highest-rated restaurants?
4. Which cuisines are most popular?
5. What is the relationship between cost and ratings?

Step 4: Data Visualization

Professional visualizations were created using Matplotlib and Seaborn.

Step 5: Machine Learning

A Random Forest Regressor was trained to predict restaurant ratings using selected features.

6. Exploratory Data Analysis Results

- **Online Delivery vs Rating:** Analysis showed that restaurants offering online delivery generally receive better customer engagement and competitive ratings.
- **Table Booking vs Rating:** Restaurants providing table booking facilities tend to have higher average ratings.
- **City-wise Rating Analysis:** Several cities consistently demonstrate better restaurant performance and customer satisfaction.
- **Cuisine Analysis:** Popular cuisines dominate customer preferences and restaurant distribution.
- **Cost vs Rating Analysis:** Higher prices do not necessarily guarantee higher ratings.

7. Visualizations

The following visualizations were generated:

1. Online Delivery vs Rating
2. Table Booking vs Rating

3. Top Rated Cities
4. Top Cuisines
5. Cost vs Rating Scatter Plot
6. Votes vs Rating Scatter Plot
7. Correlation Heatmap
8. Feature Importance Chart

8. Machine Learning Model

Model Used

Random Forest Regressor

Features Used

- Average Cost for Two
- Price Range
- Votes
- Has Online Delivery
- Has Table Booking

Target Variable

- Aggregate Rating

Evaluation Metrics

- R^2 Score
- Mean Absolute Error (MAE)

The model was trained and evaluated to predict restaurant ratings based on business features.

9. Business Insights

1. Customer votes strongly influence restaurant ratings.
2. Online delivery services improve customer engagement.
3. Table booking facilities are associated with higher ratings.
4. Restaurant performance varies across different cities.
5. Customer satisfaction depends more on service quality than pricing.

10. Recommendations

- **Recommendation 1:** Expand online delivery services to improve customer reach and convenience.
- **Recommendation 2:** Implement reservation systems to enhance customer experience.
- **Recommendation 3:** Focus on service quality improvements rather than increasing prices.
- **Recommendation 4:** Target expansion opportunities in cities with strong restaurant performance.
- **Recommendation 5:** Encourage customer reviews and feedback to increase engagement.

11. Conclusion

This project successfully analysed the Zomato restaurant dataset and identified key factors influencing restaurant ratings. The analysis demonstrated that customer engagement, online delivery services, and table booking facilities play significant roles in restaurant success. The machine learning model provided additional predictive insights, helping businesses make informed decisions based on data.